Smart Warehouse Safety: Computer Vision for Forklift Driver Monitoring in Warehouse Setting

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Abstract*—Safety for forklifts and warehouse workers must be prioritized. Our study introduces a computer vision-based solution for promptly recognizing unsafe behaviors in real time using deep learning and pose estimation. The primary goal is to identify unsafe actions of forklift drivers, enhance workplace safety, and help managers spot dangerous trends or patterns early on. The research will integrate Convolutional Neural Networks and Pose Estimation. Specifically, we will use the "You Only Look Once" architecture version 8 of the CNN model for object detection to identify forklift drivers. Subsequently, the YOLO version 8 pose will detect and estimate forklift driver body joints. Our findings indicate that this approach achieves a detection accuracy rate of over 91 percent and an f1 score of the pose estimation model exceeding 0.96. Importantly, the potential of our solution is not limited to forklift safety in warehouses. Future work will focus on adapting the model to different warehouse settings and other high-risk industries, demonstrating the versatility and wide-ranging applicability of our research.

Keywords*—computer vision, object detection, pose estimation, forklift safety monitoring

I. Introduction

A forklift is a motorized machine for loading, unloading, and transporting items. It has a fork for lifting and a mast for moving it up and down [1]. Due to their versatility, ease of operation, and high demand in material handling, forklift trucks have become widely used in warehouses, manufacturing plants, and construction sites.

Occupational Safety and Health Administration (OSHA) reports 34,900 serious forklift accidents and 61,800 non-serious injuries related to forklift accidents annually in the United States [2]. From 2022 to 2023, OSHA recorded 205 accidents involving forklifts, with 128 resulting in fatalities. The Japan International Center for Occupational Safety and Health (JICOSH) also reports that nearly 2,000 workers die annually from industrial accidents in the 2000s [3]. The causes of these accidents, which are largely preventable, include limited driver awareness, collisions, rollovers, sudden or excessive movement, reduced visibility for forklift operators, falls from forklifts, dropping of loads, and collisions with stacked loads. These accidents are often attributed to a lack of forklift awareness and insufficient safety training, significantly increasing the risk of collisions and operational accidents. These statistics underscore the urgent need for innovative solutions in forklift safety, making our research all the more crucial.

Integrating technological solutions into forklifts and warehouses is crucial for improving safety, as many accidents result from human error [4]. Utilizing image detection systems and pose estimation solutions allows for accurate tracking of operators, helping to mitigate risky situations. Therefore, it is recommended that a monitoring system be implemented to assist managers and supervisors in identifying and addressing unsafe trends or patterns in warehouse forklift operations.

In this study, the main benefit of the proposed framework is to enhance the safety and efficiency of forklift operations within a warehouse setting. The contributions of the work can be summarized as follows:

- Gather and collect data from the existing CCTV camera to train an Object Detection Model and Pose Estimation Model for warehouse settings.
- Develop a Pose Estimation Model to identify the unsafe behavior of the forklift drivers.
- Evaluate the custom model using k-fold cross-validation techniques and classification reports. The proposed method obtains over 91 percent detection performance and 0.96 F1-Score for pose estimation.

II. Preliminaries

A. You Only Looks Once

Redmon and colleagues proposed a faster image recognition scheme called You Only Look Once, or YOLO, in 2015 [5]. YOLO speeds up the typical CNN learning style by performing a one-stage scan over the image and transforming an image classification problem into logistic regression learning

Ultralytics released YOLO version 8 [6] in January 2023. The architecture uses a similar backbone as YOLOv5 with some changes on the CSPLayer called the C2f module. Also, YOLOv8 utilizes an anchor-free detection head to quickly detect and classify objects in images. The C2f module (cross-stage partial bottleneck with two convolutions) combines high-level features with contextual information to improve detection accuracy.

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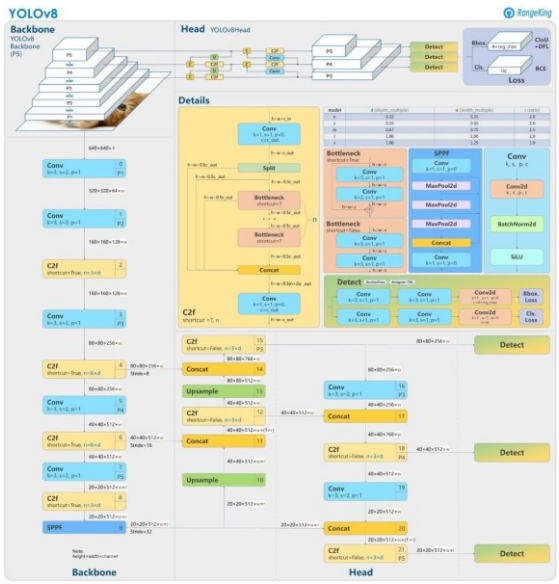


Fig. 1. YOLO version 8 Model Structure [6].

B. Pose Estimation Models based on YOLO version 8

In the YOLOv8 series, there are six pose estimation models [7], including YOLOv8s-pose, YOLOv8n-pose, YOLOv8m-pose, YOLOv8l-pose, YOLOv8x-pose, and YOLOv8x-pose-p6 that add the functionality of human pose estimation to the YOLOv8 model. Traditional 2D human pose estimation methods are mainly classified as bottom-up and top-down. Top-down methods [8] first detect all human positions in the image and then independently estimate each detected person's pose information and size. Human body posture estimation must judge and locate the target and interpret the target's rotation angle in three-dimensional space. The YOLOv8n-pose output contains the coordinates of 17 key points with two axes (x-axis and y-axis) [9], as shown in Figure 2.

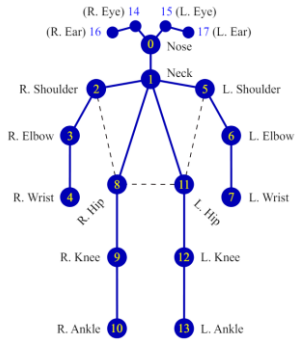


Fig. 2. COCO keypoint output format [9].

C. Related work

Accidents with forklifts often result in serious injuries and complications. The literature review [4] found that training, education, the Internet of Things, and work measurements can positively impact productivity. Further research is needed on integrating computer vision technology to gather real-time data on the operator's movements and interactions with forklifts.

The use of Deep Learning for forklift safety in Thailand [10] showed 84% success in detecting objects near blind spots. The study suggested using YOLO for detection and a Liquid Crystal Display (LCD) for visual alerts. It was noted that the current display only allows for a front or rear camera view at a time.

In [11], the author discusses using computer vision to detect dangerous behaviors and promote safety in manufacturing. The study implemented YOLO version 3 with a self-made dataset to develop object and hand detection using Mediapipe. A camera was installed onboard a forklift to film all the operators' movements and identify unsafe behavior.

Another study [12] proposed that cameras could be installed on forklifts to monitor operators and improve safety at construction sites. The system involves three cameras placed on the forklift, warehouse, and pallet to use advanced computer vision technology to enhance safety. However, there are certain limitations that necessitate careful consideration and foster discussion on technology, cost, safety, and compliance.

D. CiRA Core IDE

CiRA Core [13] is a visual programming platform built on the Robot Operating System Framework (ROS) by the College of Advanced Manufacturing Innovation at King Mongkut's Institute of Technology, Ladkrabang. The IDE runs ROS in the background and integrates it with the Darknet framework. It offers fast installation, supports CPU and GPU computation, and provides a user-friendly environment. The integrated development environment (IDE) supports basic block-based coding, computer vision tools, and external connectivity such as the Internet of Things (IoT) and Serial Communication.

III. Propose Framework

The proposed framework utilizes computer vision to monitor safety and detect unsafe behavior among forklift drivers. It comprises three integrated modules: Object Detection, Pose Estimation, and Monitoring System. These modules operate within a central server, processing and storing data in a local database. The system is designed to notify workers when an unsafe action has occurred. It leverages existing warehouse cameras to stream real-time images to the server for processing by the Object Detection and Pose Estimation modules. Subsequently, the Monitoring System module assesses the results for unsafe behavior, as depicted in Figure 3.

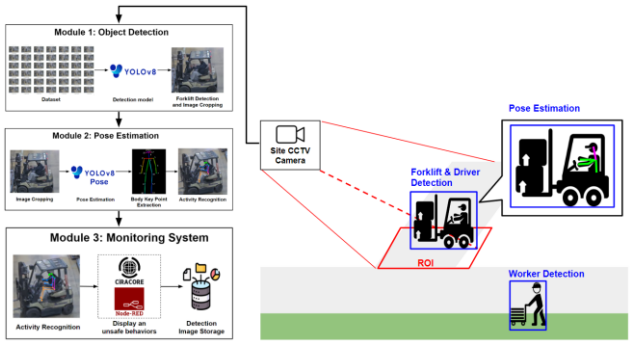


Fig. 3. The Proposed Framework Diagram.

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A. Module 1: Object Detect-based YOLOv8

Our first module focuses on creating a deep-learning model to categorize images of workers, forklift trucks, and drivers. We use the YOLOv8 Architecture to process data from the warehouse's CCTV camera for object localization. Our dataset includes 5,417 images, each of which will be individually labeled to generate an annotation file. The dataset is segmented into Train (80%), Validation (10%), and Test (10%) Datasets. We are using the automatic labeling tool called GTGenerator from CiRA Core IDE to streamline the labeling process. Nevertheless, we must carefully review and correct each annotation file to ensure the accuracy of the labeled images for the training phase.

We are working on training an accurate and efficient object detection model by adjusting specific parameters. Before starting the model training, the input image size is set to 640 pixels for width and height. The batch size determines the number of images used to update the weights in each process, which is set to 16. Additionally, we use subdivisions to handle a fraction of the batch size on the GPU, with a value of 8. The learning rate is set at 0.00261, with a momentum of 0.9 and decay of 0.0005. The model has undergone 5,000 iterations.

We evaluated our custom object detection model using the K-fold cross-validation technique. Our dataset was split into five groups. Then, we trained the model five times using different groups as the test dataset and the remaining ones as the training dataset. After training, we calculated each model's precision, recall, and average precision and averaged the results. Precision and recall were employed to evaluate the model. Precision measures the classifier's ability to accurately identify negative instances, while recall measures its ability to identify all positive instances. Additionally, we used a test image dataset to evaluate the model's accuracy with a 50 percent threshold. The results of our k-fold cross-validation analysis are provided in Table I in section IV, titled "Evaluation and Results." Lastly, we used Regions of Interest (ROI) to locate where forklift trucks and workers operate to improve our system's performance and avoid misdetection.

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

$$Accuracy\ \% = \left(\frac{Number\ of\ Correctly\ Detection\ Images}{Total\ Number\ of\ Test\ Images}\right) \times 100 \quad (3)$$

$$Error\ = \frac{1}{k}\sum_{i=1}^{k} x_i \quad (4)$$

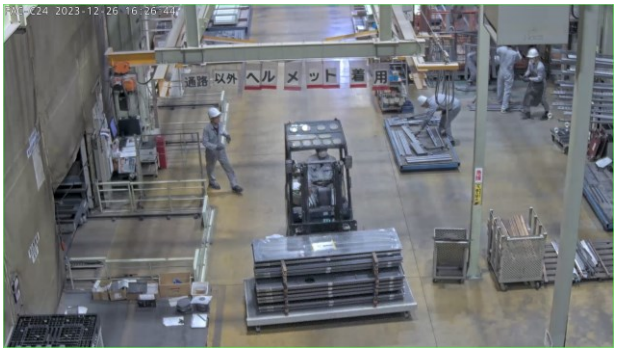


Fig. 4. A single top-down perspective image.

B. Module 2: Pose Estimation-based YOLOv8 pose

The study referenced in citation [7] suggests using YOLOv8n-pose for applications that require fast inference speed. In order to work with CiRA Core IDE, we used Python to import the Pose Estimation Model and extract the key points of the human body. The model provides the coordinates of these points along the x and y axes. We expanded our limited dataset using image augmentation, which involved rotating the images by 30 degrees. This increased to 2,600 images of forklift drivers holding the wheel and 624 images of forklift drivers releasing the wheel. We then utilized the YOLOv8n-Pose model to extract key points from the forklift driver and save them in CSV file format. The dataset consists of 1,340 rows, with 963 holding rows and 377 releasing rows, each containing pose labels and 17 body key points, resulting in 34 coordinate values (x and y axes for each human body key point). Preprocessing the key points dataset involved handling missing values and labeling the rows. We padded the dataset with zero values to ensure a uniform data structure for the model to handle it more easily. The second step involved labeling the rows based on the image augmentation dataset used.

	A	B	C	D	E	F	G	H	I
1	pose	nose_x	nose_y	lEye_x	lEye_y	rEye_x	rEye_y	lEar_x	lEar_y
955	holding	0.44312	0.3101	0	0	0.441592	0.297026	0	0
956	holding	0.551	0.302482	0.558605	0.289997	0	0	0.58489	0.294565
957	holding	0.512026	0.322309	0.518349	0.310047	0	0	0.547503	0.313719
958	holding	0	0	0	0	0	0	0.53943	0.328039
959	holding	0.484141	0.34791	0.486468	0.336642	0	0	0.515097	0.330667
960	holding	0.508204	0.349784	0.511073	0.34042	0	0	0.532756	0.33764
961	holding	0.478454	0.329009	0.484354	0.318059	0	0	0.509104	0.320922
962	holding	0	0	0.493652	0.345492	0	0	0.513045	0.340897
963	holding	0.475837	0.329581	0.481055	0.318395	0	0	0.505745	0.312324
964	holding	0.4612703	0.3388208	0.4652823	0.3276286	0.0	0.0	0.4883321	0.3179185
965	releasing	0.578298	0.307282	0.583222	0.294472	0	0	0.614994	0.296433
966	releasing	0.555852	0.306437	0.562226	0.291569	0	0	0.589894	0.290752
967	releasing	0.588781	0.320011	0.595337	0.306859	0	0	0.631887	0.303806
968	releasing	0.560935	0.323089	0.568426	0.308956	0	0	0.597067	0.305066
969	releasing	0.506745	0.318499	0.510794	0.306413	0	0	0.536953	0.306718
970	releasing	0.572066	0.319963	0.584049	0.304392	0.566416	0.309877	0.613833	0.303031
971	releasing	0.574816	0.30873	0.578829	0.297046	0	0	0.610108	0.303203
972	releasing	0.546893	0.305309	0.551935	0.292128	0	0	0.583679	0.293824
973	releasing	0.570554	0.294669	0.574731	0.282805	0	0	0.607902	0.290471
974	releasing	0.548027	0.320469	0.551228	0.307376	0	0	0.575829	0.303241

Fig. 5. Part of Forklift Driver Key Point Dataset.

To address the imbalanced dataset issue, we considered using the Synthetic Minority Oversampling Technique (SMOTE) [14]. This technique generates synthetic samples for the minority class. After employing the SMOTE technique, the dataset increased to 1,926 data points, which were then divided into 80% for the Train Dataset (1,540 data), 10% for the Validation Dataset (193 data), and 10% for the Test Dataset (193 data). With the clean and prepared dataset, we proceeded with training and testing for comparison using various architectures.

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Five architectures were considered and evaluated for training and testing. The architectural description used in this work is presented below.

Architecture 1 (LSTM + LSTM + LSTM + Dense + Dense): This architecture was influenced by the "Action Detection for Sign Language" GitHub repository [15] and employed three LSTM cell states. The layers consist of 64 neurons, 128 neurons, and 64 neurons, respectively. Subsequently, two dense layers are utilized to generate the model's predictions. This model is specifically tailored to recognize sign language actions from webcam videos by leveraging Google Mediapipe to extract key points from the face, hands, and body.

Architecture 2 (LSTM + Dropout + LSTM + Dense + Dense): This architecture consists of an LSTM layer with Dropout, followed by two dense layers. It was inspired by the model proposed in the paper "Proposal of a Real-time American Sign Language Detector using MediaPipe and Recurrent Neural Network" [16], which utilized two states of an LSTM cell: the cell state and the hidden state. In their model, the outer layer had a dimensionality of 256, and the inner layer had a dimensionality of 128. A dropout layer was used to prevent overfitting, followed by a flattened layer to shape the LSTM as a one-dimensional input. Finally, a Dense layer, a fully connected layer, was used. When evaluated with their test dataset, this network achieved an accuracy of 94%.

Architecture 3 (LSTM + LSTM + Dense + Dense): This architecture was derived from the model and findings presented in the paper titled "Human Action Recognition (HAR) Classification Using MediaPipe and Long Short-Term Memory (LSTM)" [17]. The model consisted of 2 LSTM layers and 2 Dense layers. Other hyperparameter settings included a learning rate of 0.0001 and a batch size of 4. This architecture demonstrated that the network achieved an accuracy of 95.19% compared to another model discussed in the study.

Architecture 4 (Conv1D + Conv1D + Dropout + Max Pooling + Flatten + Dense + Dense): This architectural design was sourced from the GitHub repository "ETFA Workshop 5 Hands-on Deep Learning for Industrial Applications" [18]. Unlike the previous architectures, this network incorporates convolutional layers. The architecture consists of two 1D convolutional layers with 64 neurons, a dropout layer to mitigate overfitting, a max pooling layer to reduce dimensionality, and a final section of flattened layers and two fully connected layers. The MaxPooling1D layer effectively reduces the sequence length by half. However, adjusting the dataset to accommodate the 64 neurons in the 1D convolutional layers is important. We have a total dataset of 481 samples, with 384 samples allocated for training. Each sample represents a window of the time series data, comprising 4-time steps and 34 variables or features. Convolutional networks can be particularly effective for time series analysis, as they excel in extracting features from input data, helping to identify patterns and trends. Additionally, convolution enables data downsampling, which decreases the computational complexity of the model, making it more manageable for training and execution. This network was initially assessed using the UCI-HAR dataset and achieved an accuracy of 89.89%.

Architecture 5 (Conv1D + Conv1D + Max Pooling + Bidirectional LSTM + Dropout + Dense + Dense): This architecture includes two Conv1D layers, a Max Pooling layer, a Bidirectional LSTM layer, a Dropout layer, and two Dense layers. The code was influenced by a study named "Wearable Sensor-Based Human Activity Recognition with Hybrid Deep Learning Model" [19], which proposed the use of 1D convolution and bidirectional LSTM for human activity recognition (HAR). By utilizing convolutional layers at the beginning of the architecture and integrating bidirectional LSTM, the network can capture information from both preceding and subsequent inputs, leading to a better comprehension of dependencies and relationships within the sequence. The testing dataset used for evaluation aligns with that of Architecture 7. Upon evaluation with the UCI-HAR dataset, this network achieved an accuracy of 95.48%.

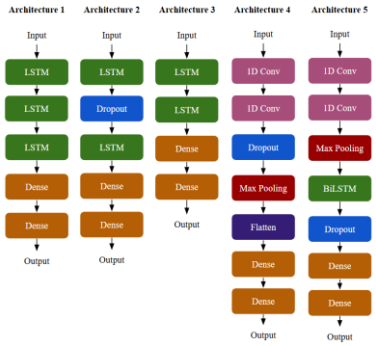


Fig. 6. Five Architecture Diagrams.

We are compiling a classification report on the predicted values, including precision, recall, and F1 score. The F1 score represents the weighted harmonic mean of precision and recall, taking into account the best and worst possible scores of 1.0 and 0.0, respectively. When comparing classification models, it is advisable to use the weighted average of the F1 score rather than the accuracy percentage. The F1 score can be calculated using the following equation (5).

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{5}$$

C. *Module 3: Monitoring System*

To monitor the unsafe behavior, we utilized the CiRA Core IDE to display detected information on a computer monitor. A red indicates an unsafe activity, whereas a green indicates a safe activity. When unsafe activities are detected, the data is captured in images and saved as JPG files in local storage. Regions of Interest (ROI) are used in this section to identify specific areas where workers and forklift trucks were likely present. This approach helped to prevent false detections and enhance the system's performance. Additionally, we develop a Node-RED UI to showcase real-time streaming camera results, detection outputs, and indicators of unsafe activity. The monitoring of CiRA Core Workspace and Node-RED UI, as illustrated in Figure 7, Figure 8, and Figure 9.

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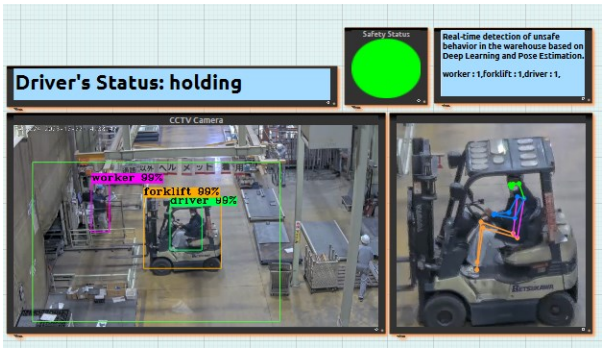


Fig. 7. Safe Activity Indicates on CiRA Core Workspace.

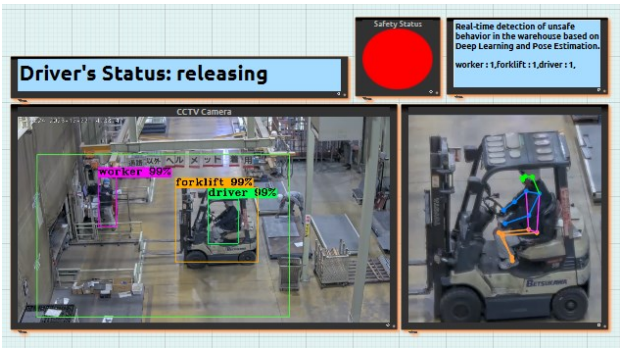


Fig. 8. Unsafe Activity Indicates CiRA Core Workspace.

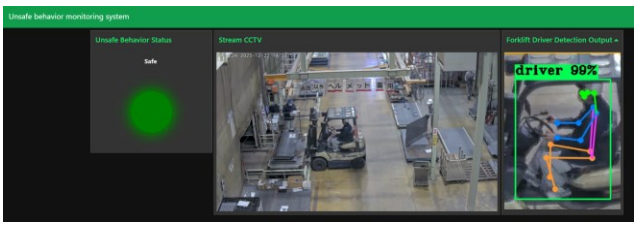


Fig. 9. Node-RED UI Monitoring.

IV. Evaluation and Results

The system was successfully trained and evaluated on a Windows PC with 16 GB of RAM, a 64-bit Operating System, an Intel(R) Core(TM) i7-12700H CPU, and an Nvidia GeForce RTX 3070 8 GB graphics processing unit (GPU). The upcoming section will discuss the results of the object detection model evaluation and pose estimation model evaluation.

A. Object Detection Model Evaluation

In our Object Detection Model, we conducted a quantitative comparison with the state-of-the-art YOLOv8 model, which we customized for our dataset. The results show that YOLOv8, with the finetune parameter, effectively identifies forklift trucks, drivers, and workers. We determined the best model for our custom dataset using the K-Fold Cross-Validation method. The highest accuracy achieved was 99.76% in the 5th K-Fold dataset. The average precision, recall, and accuracy are 0.913, 0.975, and 97.76%, respectively.

TABLE I. Cross Validation (K = 5)

	x_1	x_2	x_3	x_4	x_5	E
Precision	0.931	0.926	0.914	0.876	0.921	0.913
Recall	0.978	0.939	0.991	0.982	0.984	0.975
Accuracy%	99.30	91.45	98.84	99.53	99.76	97.76

B. Pose Estimation Model Evaluation

The report contains the classification results from our five models. The trained and weighted models were saved locally as .h5 files, using the highest validation categorical accuracy. In the pose estimation classification reports, all five architecture models achieved F1 scores ranging from 0.96 to 0.99. Architecture 2 demonstrated the highest testing accuracy, while Architecture 4 showed the lowest. As detailed in Table II, it is important to prioritize the F1 score over accuracy, especially in unbalanced problems.

TABLE II. Five Architectures Classification Report

	Epoch	Train Accuracy	Test Accuracy	F1 Score
Architecture 1	26	0.9942	0.9845	0.98
Architecture 2	50	0.9968	0.9896	0.99
Architecture 3	17	0.9974	0.9689	0.97
Architecture 4	70	0.9948	0.9592	0.96
Architecture 5	170	0.9896	0.9796	0.98

V. Conclusion and Future Work

The paper presents a proposed framework designed to identify unsafe behaviors among forklift drivers in warehouse settings. The proposed system utilizes the existing CCTV camera infrastructure to automatically detect driver behavior and enhance workplace safety. It leverages advanced object detection and human pose detection technologies. The system demonstrates high accuracy in detecting forklift trucks, drivers, and workers, achieving a rate of over 91%. However, concerns persist regarding the accuracy of the driver's key points, as they depend on the performance of the detection model. The system's experiments indicate that Architecture 2 yielded the highest F1 score for the approach utilized in Module 2: Pose Estimation. Nevertheless, Implementation of the solution requires an initial investment and approval from management for the training and testing phase.

Future research will focus on quickly notifying or predicting accidents in the warehouse. We will also work on implementing a model to detect the human body, which will help identify key points more accurately. Additionally, we plan to train a neural network to better recognize the activities and poses of workers or operators in the warehouse.

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